0019. Remove Nth Node From End of List

Category: Linked List | LeetCode: https://leetcode.com/problems/remove-nth-no

Problem

- Remove the n-th node from the end of the list and return the head, ideally in one pass.
- Example: 1,2,3,4,5 with n = 2 → 1,2,3,5.

Intuition

- Keep two pointers n nodes apart. When the front pointer falls off the end, the back pointer sits just before the node to delete.
- Starting from a dummy node handles deleting the head.

Brute Force

- Count the length L, then walk L - n steps: two passes.

Optimal Approach

- Dummy.next = head; fast = slow = dummy.
- Move fast n+1 steps.
- Move both until fast is null; slow.next is the target.
- Slow.next = slow.next.next.

Worked Example

```
1,2,3,4,5, n = 2.

fast moves 3 steps: dummy -> 1 -> 2 -> 3
move both: (slow=1,fast=4) (2,5) (3,null)
slow=3 -> remove 4 -> 1,2,3,5
```

Java Solution

```java
public class RemoveNthNodeFromEnd {
    public ListNode removeNthFromEnd(ListNode head, int n) {
        ListNode dummy = new ListNode(0, head), fast = dummy, slow = dummy;
        for (int i = 0; i <= n; i++) fast = fast.next;
        while (fast != null) { fast = fast.next; slow = slow.next; }
        slow.next = slow.next.next;
        return dummy.next;
    }
}
```

Complexity

- Time: O(L). Space: O(1).

Pitfalls

- Removing the head (n == length) is why the dummy node exists.